



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
SECOND SESSIONAL
SUBJECT: (CE-514) ADVANCED ALGORITHMS

Examination : B.Tech Semester-V
Date : 07/09/2022
Time : 12:30 P.M. - 01:45 P.M.

Seat No : _____
Day : Wednesday
Max. Marks : 36

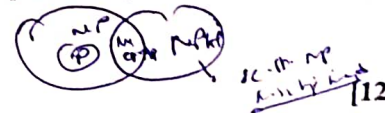
INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed.

[12]

- CO1 U (a) State true or false and justify your answer, "All NP Hard Problems are optimization problems." [2]
- CO2 R (b) Give an example of NP Complete problem. [2]
- CO2 A (c) Write a non deterministic algorithm to solve the 3 SAT problem. [2]
- CO2 U (d) How Horspool's algorithm is different and better than the Naive algorithm? Explain with a suitable example. [2]
- CO1 A (e) Assume Pattern P = "364", Modulus value $q=11$, and Text T = "11122364133". How many spurious and valid hits are there assuming radix 10? Show your computation. [2]
- CO2 N (f) How many persons are required in the room so that the probability of at least two persons having the same birthday is 0.75? [2]



Q.2 Attempt Any Two of the following questions.

[12]

- CO2 U (a) (i) What is the basic Idea of the KMP string Matching algorithm? How is it better than the Naive one? [1]
- A (ii) Trace the algorithm to create the LPS array using P = "ABCABCA" as input. [2]
- A (iii) Trace the algorithm for KMP Pattern Matching over Text T = "ACCABCABCABCA". Show every necessary step. [3]
- CO1 C (b) (i) Draw/Create a Finite Automaton and transition table for the following Pattern: GCGCGAG, given the alphabet set $\sigma = \{A, C, T, G\}$. [4]
- A (ii) Using that transition table, do pattern matching over the Text: ACACGCGCGCGAG. [2]
- CO2 R (c) (i) Write an algorithm to check whether two line segments P1P2 and P3P4 intersect or not. [3]
- N (ii) Consider the following set of Points: P1=(2,8) P2=(7,4) P3=(-4,3) P4=(10,10). Apply the Line segment Intersection algorithm over it and check if they intersect or not. Show every step of Computation. [3]

Q.3 Attempt the following questions.

[12]

- CO1 A (a) Prove that the problem of finding the Minimum set cover is NP Complete. [6]
- CO2 U (b) Prove that the problem to find out Minimum Vertex Cover from a graph is polynomially equivalent to finding the Largest Independent set in a graph. [6]

OR

Q.3 Attempt the following questions.

[12]

- CO1 C (a) Prove or give a counterexample that the 2-SAT problem is reducible to 3-SAT problem. [6]
- CO2 A (b) Prove that the 3 coloring of a graph is NP - Hard. [6]